

On the Practical Impacts of Inter-Modal-Interference (IMI) in Hollow-Core Fibers for High-Speed Data Center Interconnects

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Abstract We measured IMI-induced SNR after different HCF transmission with different transponders and frequency and found that SNR can be worse than theoretical value. A delivered OSNR model validated by 1.6T/λ transmission experiments identify -54dB/km as a practical IMI threshold, providing guidance on HCF-based DCI standardization/optimization. ©2026 The Author(s)

Introduction

Hollow-core fiber (HCF) has emerged as a transformative technology for next-generation optical networks, with recent advances achieving record-low attenuation [1–4] and inherently offering low nonlinearity and 30% latency reduction compared to standard single-mode fiber (SSMF). These advantages make HCF attractive for latency-sensitive datacenter interconnect (DCI), where hyperscale operators are already exploring deployment [5–8]. The key design question is therefore how HCF-specific impairments translate into reliable quality-of-transmission (QoT) margins.

Two impairments matter most for HCF QoT modeling: gas line absorption (GLA) and inter-modal interference (IMI). GLAs introduce filtering-like spectral penalties in the L-band, though mitigation strategies have been demonstrated [9–11]. IMI, arising from the excitation of higher-order modes (HOMs) in intrinsically multimodal HCFs [12, 13], acts as additional noise source that degrades QoT, ultimately limiting achievable network capacity and reach [14–17]. Theoretical and simulation study therefore specify IMI coefficient: below -60dB/km to avoid significant QoT degradation [12].

However, two critical gaps remain between these theoretical insights and practical network deployment. First, while state-of-the-art (SOTA) HCFs have demonstrated -74dB/km IMI under laboratory conditions [3, 14], large-scale production variability, field splicing and degradation inevitably worsen the IMI after deployment [3, 16]. Second, existing studies on IMI impact at the network level are predominantly simulation-based [15–17]. Experimental validation of IMI-induced penalties on commercial transponders remains scarce. Yet, DCI networks impose stringent requirements on high-speed transponders operating at high spectral efficiency (SE) with tight margins [18], making it imperative to connect IMI measurements to QoT prediction and to define how much IMI is acceptable for reliable HCF DCI network deployment.

In this paper, we experimentally measured the

IMI-induced SNR penalties on commercial coherent transponders (800G to 1.6T, 8 bit/s/Hz SE) using three HCFs with different IMI levels. We compare theoretical SNR derived from measured IMI coefficients against actual transmission penalties, revealing that practical penalties can exceed theory and vary with frequencies and different transponders. Based on experimental results, we build a QoT model for HCF-based DCI scenario that incorporates a 2-dB uncertainty margin on IMI-induced SNR, validated by 1.6T/λ WDM HCF transmission experiments with high power EDFA. The results identify -54dB/km as the acceptable IMI coefficient threshold for 1.6T/λ HCF DCI transmission, providing essential guidance for HCF-based DCI network modeling, standardization, and optimization.

IMI measurement and SNR calculation

Tab. 1 details the three HCFs under study, with IMI coefficients measured at 1550nm using two methods [2, 19, 20]: (i) time method: recording power variations over 10 minutes, and (ii) swept-wavelength method: measuring loss spectra from 1550 to 1555nm at 0.1pm resolution. Both methods yield consistent IMI coefficients, expressed in dB/km as the noise-to-signal power ratio per unit length [2, 13, 15, 20]. The loss spectra of three HCFs at 0.1 pm resolution are shown in Fig. 1(b) where spectral thickness provides a qualitative indication of IMI level. For X km HCF with IMI coefficient Y dB/km, the theoretical IMI-induced SNR [15] is

$$\text{SNR}_{\text{IMI}} = -(Y + 10 \cdot \log_{10} X) \quad (1)$$

The theoretical SNR_{IMI} values in Tab. 1 are approximately 20, 25, and 29 dB for HCF-1, HCF-2, and HCF-3, respectively. Total SNR after HCF transmission treats all noise sources as independent and additive:

$$\text{SNR}_{\text{tot}}^{-1} = \text{SNR}_{\text{ASE}}^{-1} + \text{SNR}_{\text{IMI}}^{-1} + \text{SNR}_{\text{other}}^{-1} \quad (2)$$

where SNR_{ASE} is amplifier noise and $\text{SNR}_{\text{other}}$ accounts for nonlinearities, filtering and transponder penalties, however, can be negligible in short-reach HCF-based DCI scenario.

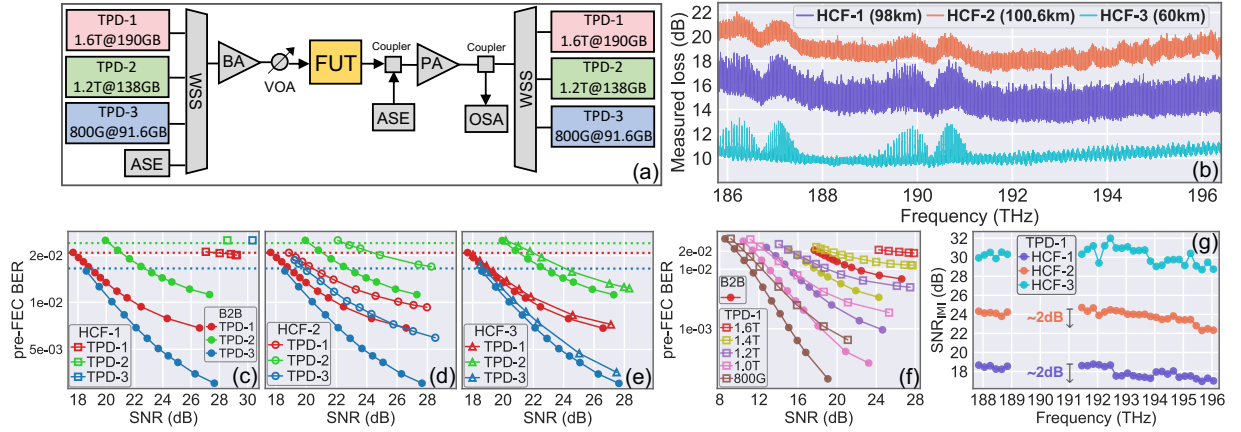


Fig. 1: (a) Experimental setup; (b) the measured loss spectra of HCF-1, -2 and -3 at 0.1 pm resolution; and the BER-SNR curves measured at B2B and measured with (c) HCF-1, (d) HCF-2, (e) HCF-3 as FUT, for different TPDs; (f) the BER-SNR curves measured at B2B and measured after HCF-1 for TPD-1 operated at different data-rates but the same 190GBaud; (g) practical SNR_{IMI} calculated from the difference of BER-SNR curves between B2B and after HCF transmission for TPD-1.

Tab. 1: Details of 3 different HCFs under study

Fiber	Length [km]	Time Method @1550nm		Wavelength Method @1550nm	
		IMI coef. [dB/km]	SNR_{IMI} [dB]	IMI coef. [dB/km]	SNR_{IMI} [dB]
HCF-1	98.0	-39.93	20.02	-40.5	20.59
HCF-2	100.6	-45.24	25.21	-44.72	24.70
HCF-3	60.0	-47.17	29.38	-46.94	29.16

Experimentally Measured IMI-induced SNR

To evaluate practical IMI impacts, we employed three commercial coherent transponders (TPDs): (i) 1.6Tbps/190GBaud TPD-1, (ii) 1.2Tbps/138GBaud TPD-2, and (iii) 800Gbps/91.6GBaud TPD-3, as shown in Fig. 1(a), all operating at 8 bit/s/Hz SE with WSS grids of 200, 150, and 100 GHz, respectively. The three HCFs from Tab. 1 serve as fiber under test (FUT). At 193.05 THz, the BER-SNR curves at back-to-back (B2B) and after different HCF transmission are shown in Fig. 1(c), (d) and (e). Practical SNR_{IMI} is derived from the discrepancy between B2B and HCF transmission curves, with results and inversely derived IMI coefficients via Eq. (1) given in Tab. 2.

From Fig. 1(c) and Tab. 2, we can see that for HCF-1 transmission, BER of TPD-2 and TPD-3 exceeds the Pre-Forward Error Correction (Pre-FEC) threshold and thus no corresponding SNR_{IMI} given in Tab. 2. The practical SNR_{IMI} of TPD-1 derived from Fig. 1(c) is also lower than the theoretical prediction given in Tab. 1. We infer that the accumulated IMI noise after HCF-1 transmission is too large and possibly exceeds the tolerance of DSP, which can hardly be used for actual transmission deployment. One possible solution is to decrease the data-rate of TPD so that the SNR requirement is relaxed and sufficient SNR margin is preserved. Fig. 1(f) shows the BER-SNR curves at B2B and after HCF-1 for TPD-1 at different data-rates while keeping the baud-rate at 190GBaud. The required SNR decreases as the data-rate de-

Tab. 2: Practical SNR_{IMI} for different TPDs.

		HCF-1	HCF-2	HCF-3
TPD-1	SNR_{IMI} [dB]	17.86	24.02	31.09
@1.6T	IMI coef. [dB/km]	-37.77	-44.04	-48.87
TPD-2	SNR_{IMI} [dB]	—	23.82	29.02
@1.2T	IMI coef. [dB/km]	—	-43.85	-46.8
TPD-3	SNR_{IMI} [dB]	—	24.12	31.46
@800G	IMI coef. [dB/km]	—	-44.15	-49.24

creases until the required SNR become very close between B2B and HCF-1 transmission at the data rate of 800Gbps, as shown in the brown circles and squares of Fig. 1(f). The practical SNR_{IMI} for 1.6T, 1.4T, 1.2T, 1.0T and 800G in Fig. 1(f) is 17.86, 17.65, 17.51, 17.43 and 17.43 dB, showing that the practical SNR_{IMI} stays nearly the same for a given TPD operating at different data-rate/SE.

For HCF-2 transmission, Tab. 2 shows a good agreement of practical $\text{SNR}_{\text{IMI}} \approx 24\text{dB}$ across different TPDs, however, slightly worse than the theoretical values of 25.21 and 24.7 dB in Tab. 1. For HCF-3 transmission, both the practical SNR_{IMI} of TPD-1 and TPD-3 are above 31 dB and higher than that of TPD-2 (29.02dB) and theoretical SNR_{IMI} in Tab. 1. Beyond measurement uncertainty and IMI-induced fading variability, one explanation is that digital subcarrier multiplexing (DSCM) may exhibit higher IMI tolerance as indicated in [21].

We further investigated wavelength-dependent IMI impact. Fig. 1(g) shows practical SNR_{IMI} of TPD-1 (1.6Tbps/190GBaud) across frequencies for each HCF, covering all 24 C-band and 6 L-band WDM channels (avoiding GLA-affected regions). A B2B baseline curve was measured at each frequency to isolate intrinsic TPD wavelength-dependency. It is evident that even for the same TPD and HCF, SNR_{IMI} varies across the C- and L-band with higher-frequency channels suffering greater penalties of about 2 dB as shown in Fig. 1(g). In general, experimentally mea-

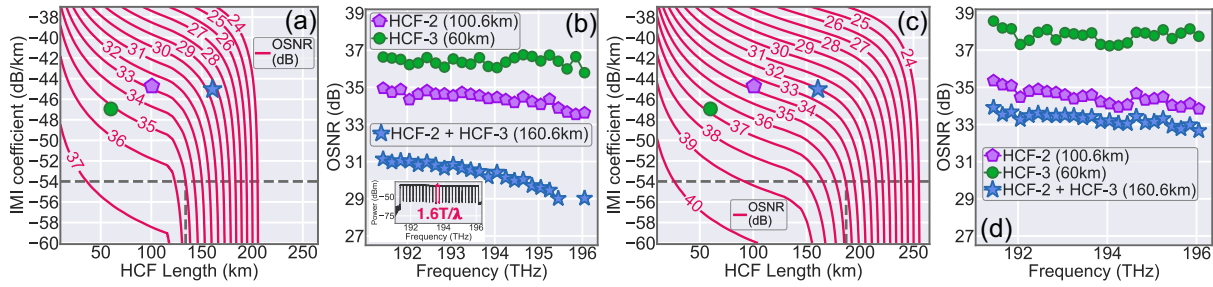


Fig. 2: (a) The delivered OSNR model with 23-dBm BA and the 3 markers correspond to (b) C-band 1.6T/λ WDM transmission experimental validations where OSNR were obtained by mapping BER from B2B curve, with HCF-2, HCF-3, and their cascade as fiber span, and the inset diagram is the OSA trace at output of BA; (c) The delivered OSNR model with 33-dBm BA and the 3 markers correspond to (d) a similar 1.6T/λ WDM transmission experimental validation with 33-dBm BA.

sured SNR_{IMI} can be slightly worse than the theoretical prediction derived from the IMI coefficient of HCF measured at 1550 nm [1, 2], due to: (i) inherent measurement uncertainty [22], (ii) wavelength-dependent IMI variation and (iii) modulation-format- and DSP-dependent tolerance to IMI [21]. Therefore, in the following section, we will build a QoT model by adding 2-dB uncertainty to theoretical SNR_{IMI} accounting for both the above factors and potential fiber degradation from field re-splicing operations.

IMI threshold characterization from OSNR Model with HP-EDFA 1.6T/λ Transmissions

We next study the minimum required IMI coefficient for unrepeated $24 \times 1.6\text{T}/190\text{GBd}/\lambda$ WDM HCF transmission in a typical DCI scenario with conventional and high-power (HP) EDFA. The delivered $\text{OSNR}@0.1\text{nm}$ model is first built using Eq.(1-2), assuming IMI coefficients measured at 1550nm. HCF loss is assumed to be 0.19dB/km, consistent with HCF-2 and HCF-3 in Fig. 1(b). Note that our model intentionally adopts conservative fiber loss parameters rather than the SOTA, ensuring worst-case operational conditions and enhancing the robustness of our conclusions. Lower-loss HCFs would improve OSNR without altering the fundamental findings.

For usual case with 23dBm booster amplifier (BA), the BA has 15-dB gain, 8-dB noise figure (NF) with a flat output profile for $24 \times 1.6\text{T}/190\text{GBaud}$ WDM channels, shown in the inset of Fig. 2(b). The pre-amplifier (PA) gain ranges from 22 to 34dB with NF from 7 to 4dB. The delivered OSNR with 23dBm-BA under varying HCF length and IMI coefficient is shown in the red lines of Fig. 2(a). To validate this model, we conducted unrepeated C-band 1.6T/λ WDM transmission experiments over HCF-2, HCF-3, and their cascade (160.6km, -45.03dB/km). OSNR at each frequency was obtained by mapping pre-FEC-BER from B2B curve, as shown in Fig. 2(b). The measured OSNR ranges are 33.5-34.9dB for HCF-2 (purple), 35.8-36.8dB for HCF-3 (green), and 29.0-31.1dB for the cascade (blue), with two frequency points exceeding the FEC threshold in the latter case. The corresponding model predic-

tions in Fig. 2(a) are around 33dB (purple), 35dB (green), and 29dB (blue), respectively, showing good agreement with measurements, particularly at higher frequencies where worst-case performance occurs. Then, we modified our model and the corresponding experimental setup by replacing the BA with 33dBm HP-EDFA and replacing PA with a low-gain-range one. In this case, BA has 25-dB gain, 9.5-dB NF and PA gain ranges from 15 to 25dB with 8dB-4dB NF. The delivered OSNR model with HP-EDFA is shown in Fig. 2(c) where the three markers with different colors correspond to the three measured curves of Fig. 2(d), showing good agreement of delivered OSNR when HP-EDFA was used, particularly at higher frequency.

Based on the OSNR model of Fig. 2(a) and (c), we identify -54dB/km as a practical IMI coefficient threshold for unrepeated 1.6T/λ HCF transmission with a required OSNR of 29dB. This is because if network operators must guarantee at least 6-dB OSNR margin to each service, then the achievable unrepeated transmission distance under -54dB/km IMI will be 134km and 187km for different EDFA case at the contour line of 35dB. Beyond 134km and 187km in Fig. 2(a) and (c) respectively, further improving IMI beyond -54dB/km yields only marginal reach gains, as performance becomes predominantly limited by EDFA-induced ASE noise. We note that the IMI threshold is largely dependent on the minimum required OSNR of the TPD, however, the commercial 1.6T coherent transponder adopted here still represents a realistic benchmark for current DCI networks.

Conclusions

We experimentally characterized the practical SNR penalties induced by IMI in HCFs with varying IMI levels using multiple commercial coherent transponders. Based on the observed uncertainty and wavelength-dependent characteristics of IMI, we built a QoT model tailored for 1.6T/λ HCF WDM transmission and validated it through experiments using both conventional and HP-EDFA. Our results establish -54dB/km as a practical IMI coefficient threshold for acceptable QoT, thus providing a practical answer to how much IMI can be accepted in HCF-based DCI deployment.

Acknowledgements

The work is supported by the Hong Kong Government Research Grants Council GRF 15225423, the Innovation and Technology Fund PRP/037/24FX, and Alibaba Innovative Research Program.

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